2. Data and AI in Private Markets

Johan Hombert

hombert@hec.fr

© Copyright 2026, Johan HOMBERT. All rights reserved

Data and AI in Finance

- Data ⇒ Prediction ⇒ Decision
- Credit markets

  Mortgages, consumer loans, business loans

  ⇒ Predict default

- Private equity

  Private equity, venture capital, project finance

  ⇒ Predict cash flow

- Insurance markets

  Health, property and casualty

  ⇒ Predict losses and damages

- Public markets (class 5)

  Asset management, market making

  ⇒ Predict future price

Road Map

Mini-Case 1: Credit Scoring

VC Game

Mini-Case 2: Zillow Offers

Statistical Underpinnings

Goodhart's Law

Discrimination and Redistributive Effects

Hirshleifer Effect

Credit Scoring with Digital Footprints: An Early Adopter

- E-commerce company in Germany, 2015[1]
- The company conducted A/B testing
  - Randomized customers viewing an item on the website
  - Treatment group: Offered the option to pay within 15 days of receiving the purchased item
  - Control group: Option not offered
  - Both groups had the same item price
- Impact
  - Control group: 45% probability of purchasing the item
  - Treated group: 85% probability of purchasing the item
- What should the company's management do?

[1] Berg, Burg, Gombovic, Puri, 2019, "On the Rise of FinTechs: Credit Scoring Using Digital Footprints" *Review of Financial Studies*

Credit Scoring with Digital Footprints

- Trade-off in offering payment facility: increased sales vs. default risk

$\Rightarrow$ Assess default risk and offer payment facility if default probability is low

- Phase 1: Purchase credit scores from a credit bureau
  - Scores based on credit history and sociodemographic data
  - Default rate: 3%
- Phase 2: Utilize proprietary customer data
  - Digital footprints: OS, email, login information, etc.
  - Does this data enhance default prediction?

Scoring Model

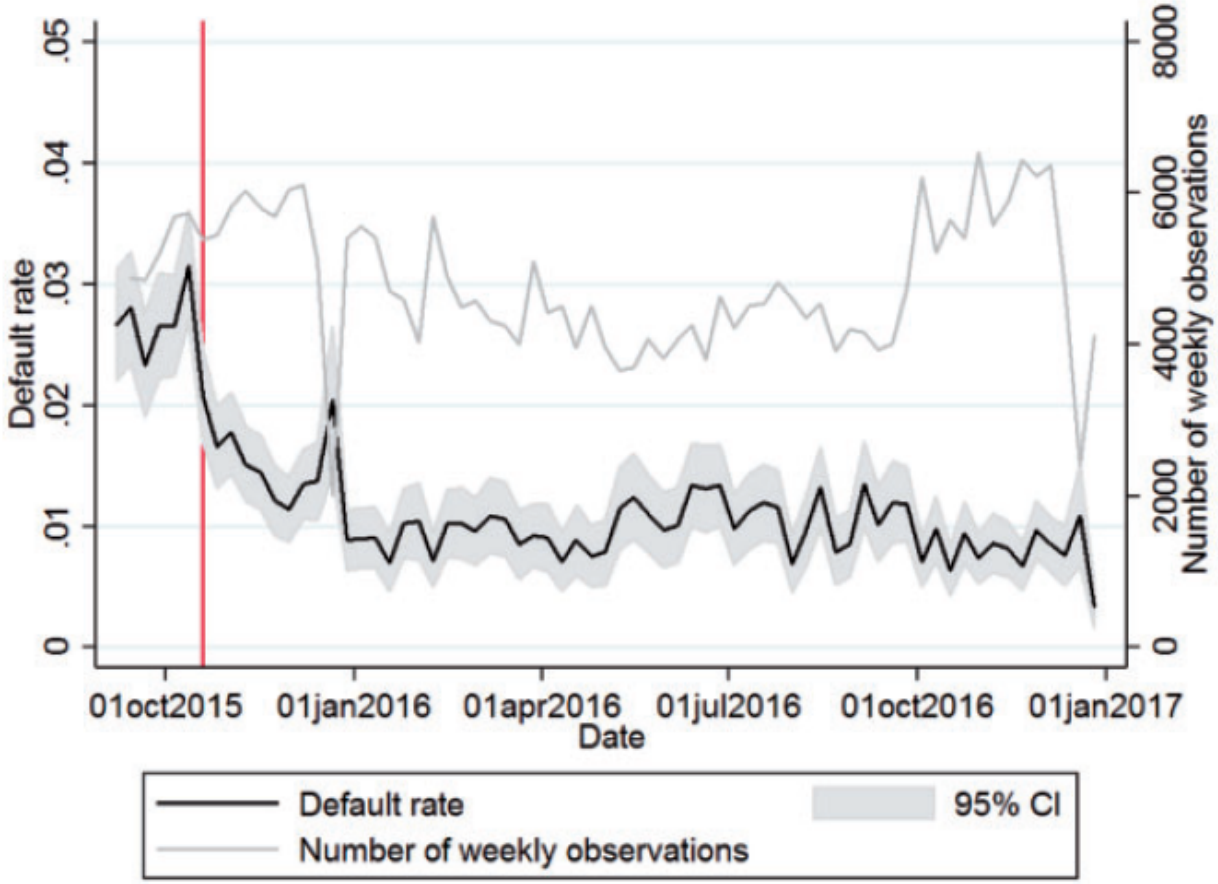
- Logistic regression to predict default
- Dependent variable: =1 if default
- Predictive variables
  - (1) Credit bureau score
  - (2) Digital footprints
  - (3) Credit bureau score + digital footprints

Default regressions (scorable customers)

	(1) Credit bureau bureau score		(2) Digital footprint		(3) Credit bureau score & digital footprint	
Variables	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
Credit bureau score	−0.17***	(−7.89)			−0.15***	(−6.67)
Device type & operating system[a]						
Desktop/Windows			Baseline		Baseline	
Desktop/Macintosh			−0.07	(−0.53)	−0.13	(−1.03)
Tablet/Android			0.29***	(3.19)	0.29***	(3.06)
Tablet/iOS			0.08	(1.05)	0.08	(0.97)
Mobile/Android			1.05***	(17.25)	0.95***	(15.34)
Mobile/iOS			0.72***	(9.07)	0.57***	(6.73)
E-mail Host [a]						
Gmx (partly paid)			Baseline		Baseline	
Web (partly paid)			0.00	(0.00)	−0.02	(−0.22)
T-Online (affluent customers)			−0.40***	(−3.90)	−0.35***	(−3.35)
Gmail (free)			0.34***	(3.81)	0.29***	(3.09)
Yahoo (free, older service)			0.75***	(9.19)	0.72***	(8.98)
Hotmail (free, older service)			0.35***	(3.70)	0.28***	(2.72)
Channel						
Paid			Baseline		Baseline	
Affiliate			−0.49***	(−5.35)	−0.54***	(−5.58)
Direct			−0.27***	(−4.25)	−0.28***	(−4.44)
Organic			−0.15*	(−1.79)	−0.15*	(−1.74)
Other			−0.47***	(−4.50)	−0.48***	(−4.36)
Checkout time						
Evening (6 p.m.-midnight)			Baseline		Baseline	
Morning (6 a.m.-noon)			0.28***	(4.50)	0.28***	(4.60)
Afternoon (noon-6 p.m.)			0.08	(1.42)	0.08	(1.47)
Night (midnight-6 a.m.)			0.79***	(7.73)	0.75***	(7.09)
Do-not-track setting			−0.02	(−0.25)	−0.07	(−0.91)
Name in e-mail			−0.28***	(−5.67)	−0.29***	(−5.70)
Number in e-mail			0.26***	(4.50)	0.23***	(3.91)
Is lowercase			0.76***	(13.10)	0.74***	(13.20)
E-mail error			1.66***	(20.00)	1.67***	(20.36)
Constant	12.42***	(5.76)	−4.92***	(−62.87)	9.97***	(4.48)
Control for *Age*, *Gender*, *Item category*, *Loan amount*, and month and region fixed effects	No		No		No	
Observations	254,819		254,819		254,819	
Pseudo R^2	**.0244**		**.0524**		**.0717**	
AUC	**0.683**		**0.696**		**0.736**	
(SE)	**(0.006)**		**(0.006)**		**(0.005)**	
Difference to AUC=50%	**0.183*** **		**0.196*** **		**0.236*** **	
Difference AUC to (1)			**0.013***		**0.053*** **	

Impact on Default Rate

- The new credit scoring model is put in production in October 2015
- Default rate divided by 3



Data and AI in Private Markets

- Using data and AI for prediction and decision-making can be very powerful
- But also important pitfalls to avoid
  - Adverse selection
  - Goodhart's Law
  - Discrimination and redistributive effects
  - Hirshleifer effect in insurance

Road Map

Mini-Case 1: Credit Scoring

VC Game

Mini-Case 2: Zillow Offers

Statistical Underpinnings

Goodhart's Law

Discrimination and Redistributive Effects

Hirshleifer Effect

VC Game

- You manage a venture capital (VC) fund
- Potential deal: Acquire an ownership stake in a startup in exchange for cash payment P
- Discounted value of ownership stake is uncertain: V drawn from uniform distribution between 20 and 100
- You further analyze the company to refine your valuation $\Rightarrow$ your best estimate of V: check your private signal
- Your best estimate is unbiased: Generated as the true V plus Gaussian noise with mean 0 and s.d. 20

VC Game

- You compete with other VC funds (students) who also have their own estimate of V
- Each VC makes an offer P
- The highest offer wins the deal
- Your profit (or loss) is
  - $V - P$ if you win the deal at price P and the true value is V
  - zero if you don't win the deal
- Submit your bid at https://forms.gle/seYd1YQAkhLLe3WN6

Winner's Curse

- Estimates are unbiased, i.e., the average is $V \Rightarrow$ some are above, some are below $\Rightarrow$ the highest is above V

- If all bid their best estimate, the winning price is the highest estimate $\Rightarrow$ above $V \Rightarrow$ winner overpays

- This is the **winner's curse**

VC and AI

Q. What are other limitations of AI in the VC context?

VC and AI

Q. What are other limitations of AI in the VC context?

The Review of Financial Studies

Data-Driven Investors

Maxime Bonelli <== HEC PhD 2023
London Business School, United Kingdom

How does the increased use of data technologies, like machine learning, by financial intermediaries affect the allocation of capital towards innovation? I study this question in the context of startup financing by venture capitalists (VCs). While VCs adopting data technologies become better at screening startups similar to those in historical data, they tilt their investments towards this pool and become concurrently less likely to finance innovative startups that achieve rare major success. Plausibly exogenous variations in VCs' screening automation suggest that these effects are causal. These findings highlight how investors' adoption of data technologies can have real effects through innovation financing. (*JEL* G24, L26, O30)

Winner's Curse Everywhere

- **Lending:** Borrowers accept your loan offer when other lenders charge higher rates, indicating your credit risk estimate is too optimistic
- **Insurance:** Customers buy insurance from you when other insurers charge higher premiums, indicating your risk estimate is too optimistic
- **Trading:** See class 5
- Auctions, etc.
- The winner's curse is also called **adverse selection**: your offer is selected (accepted) only when it is too optimistic
- Multiple Nobel Prize-winning concept: 2001 (Akerlof, Stiglitz, Spence), 2020 (Milgrom, Wilson)

Road Map

Mini-Case 1: Credit Scoring

VC Game

Mini-Case 2: Zillow Offers

Statistical Underpinnings

Goodhart's Law

Discrimination and Redistributive Effects

Hirshleifer Effect

Zillow Offers

- Zillow founded in 2006 in Seattle by former Expedia executives
- Initial core business: digital real-estate platform
  - Home search, listings, rentals, housing information
  - Revenue from advertising, lead generation, and adjacent services (mortgages)
- in 2018 Zillow launched **Zillow Offers** = "iBuyer" service
  - Homeowners request an offer online from Zillow
  - Zillow evaluates the home (with data and AI) and makes an offer
  - If accepted, Zillow buys the home and handles closing
  - Zillow performs light repairs
  - Zillow relists and resells the home

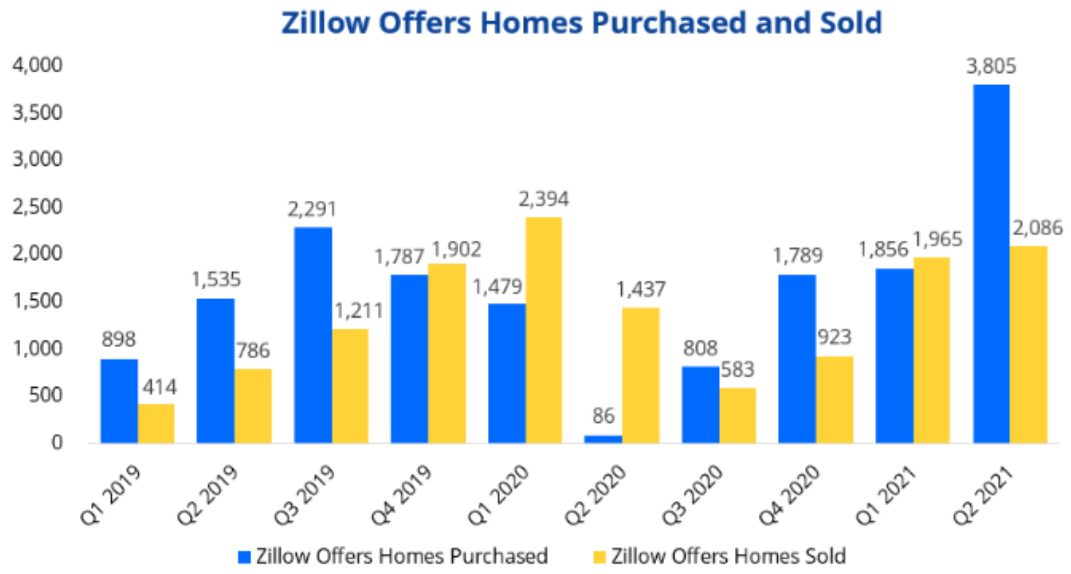
$\Rightarrow$ Zillow moved from platform intermediary to principal buyer and seller

- Value proposition to sellers: speed, certainty, and convenience (avoid repeated showings)

Estimating Home Values

- Data used
  - Home characteristics: square footage, location, photos, etc.
  - On-market data: comparable homes' listing price, days on market
  - Off-market data: tax assessments, prior sales, public records
  - Market trends: local price dynamics and seasonal demand
- Model
  - Neural network model
  - From $1 million Kaggle competition on home valuation prediction
  - Input: see above
  - Output: estimated market value of the home
- Model performance: https://www.zillow.com/zestimate/

Financial Performance



Second-Quarter 2021 Zillow Offers Unit Economics

		Total		Average per Home	% of Revenue
Homes sold		2,086			
Zillow Offers revenue	$	772,030,000	$	370,101	100.0%
Operating costs:					
Home acquisition costs*		672,592,000		322,432	87.1%
Renovation costs*		20,862,000		10,001	2.7%
Holding costs*,**		4,527,000		2,170	0.6%
Selling costs		29,544,000		14,163	3.8%
Total operating costs		727,525,000		348,766	94.2%
Return on Homes Sold Before Interest Expense[3]		44,505,000		21,335	576 bps
Interest expense*,**		3,545,000		1,699	0.5%
Return on Homes Sold After Interest Expense[3]	$	40,960,000	$	19,636	531 bps

Zillow Offers EBITDA 2021Q2: negative $29 million (negative 4% of revenue)

Q. If gross margins on homes sold are positive, why is EBITDA negative?

Financial Performance

- EBITDA 2021Q3: negative $380 million (negative 32% of revenue)

- Shareholder letter 2021Q3:

  "We recorded a $304 million write-down on inventory owned at the end of Q3 as a result of unintentionally purchasing homes at higher prices than our current estimates of future selling prices."

  "We have been able to convert only about 10% of the serious sellers who ask for a Zillow Offer."

Q. What is the connection between the inventory write-down and the 10% seller conversion rate?

Q. How to mitigate adverse selection?

(1) Bid more conservatively than the statistical estimate

- Create a margin of safety against overpaying
- But lower conversion rate

(2) Backtest after launch

- Compare forecast values with realized resale outcomes
- But reactive: losses may already be incurred

(3) Experiment with pricing on a small sample, then backtest

- Learn how offer levels affect acceptance, resale margins, and selection
- But slower scaling

(4) Focus on easy-to-value, liquid homes and markets

- Reduce hidden information and pricing error
- But weaker value proposition and narrower market

Road Map

Mini-Case 1: Credit Scoring

VC Game

Mini-Case 2: Zillow Offers

Statistical Underpinnings

Goodhart's Law

Discrimination and Redistributive Effects

Hirshleifer Effect

Statistical Underpinnings of Adverse Selection

- Highly relevant for the group work
- You make loans to businesses
- Offer interest rate R, paid at issuance (to simplify formulas)
- Cash flow (per euro lent) if loan offer is accepted:

  at issuance: $-1 + R$

  at maturity: $\begin{cases} 1 \text{ if no default} \\ 0 \text{ if default} \end{cases} = 1 - D$ where $D \in \{0, 1\}$ is default indicator

- Assume zero discount rate. Profit $= (-1 + R) + (1 - D) = R - D$
- Expected profit: NPV $= R - \mathbb{E}(D) = R - \mathbb{P}[Default]$

Statistical Underpinnings of Adverse Selection

- Default probability: $\mathbb{P}[Default] = f(X) + U$
  - X: borrower characteristics observed by lender
  - U: unobserved determinants of default risk ($\mathbb{E}[U] = 0$)
- Lender's best estimate of default probability: $f(X)$
- Suppose each loan applicant offered an interest rate R takes the loan offer if R is below a threshold: $R < \overline{R} + \mathbb{P}[Default]$, where the threshold is higher for riskier applicants
  - Rationale: Riskier applicants are more likely to accept a given offer because their chances to get a loan from another lender are lower
  - $\overline{R}$ is the maximum rate a risk-free borrower would accept
- How should the lender set the interest rate? → Simple idea: take a margin over estimated default probability: $R = f(X) + M$ with $M > 0$

Q. What is the NPV per loan offer accepted?

a. M b. more than M c. less than M d. it depends

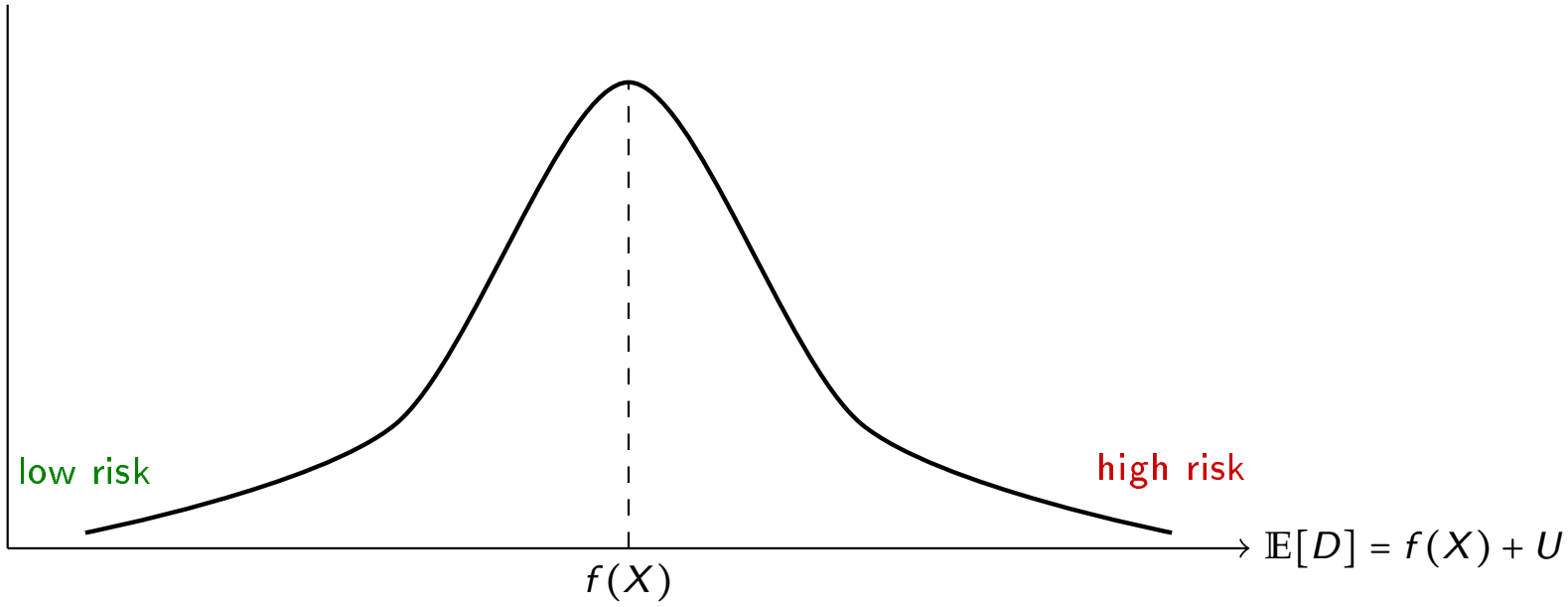
Statistical Underpinnings of Adverse Selection

- NPV per loan offer accepted

$$
\begin{aligned}
&= R - \mathbb{E}\big[D \,|\, \text{loan offer is accepted}\big] \\
&= R - \mathbb{E}\big[D \,|\, R < \overline{R} + f(X) + U\big] \\
&= f(X) + M - \mathbb{E}\big[f(X) + U \,|\, f(X) + M < \overline{R} + f(X) + U\big] \\
&= M - \mathbb{E}\big[U \,|\, U > M - \overline{R}\big] \\
&< M
\end{aligned}
$$

(You are not expected to do such calculations at the exam)

$\Rightarrow$ Adverse selection: risky applicants are more likely to accept the loan offer



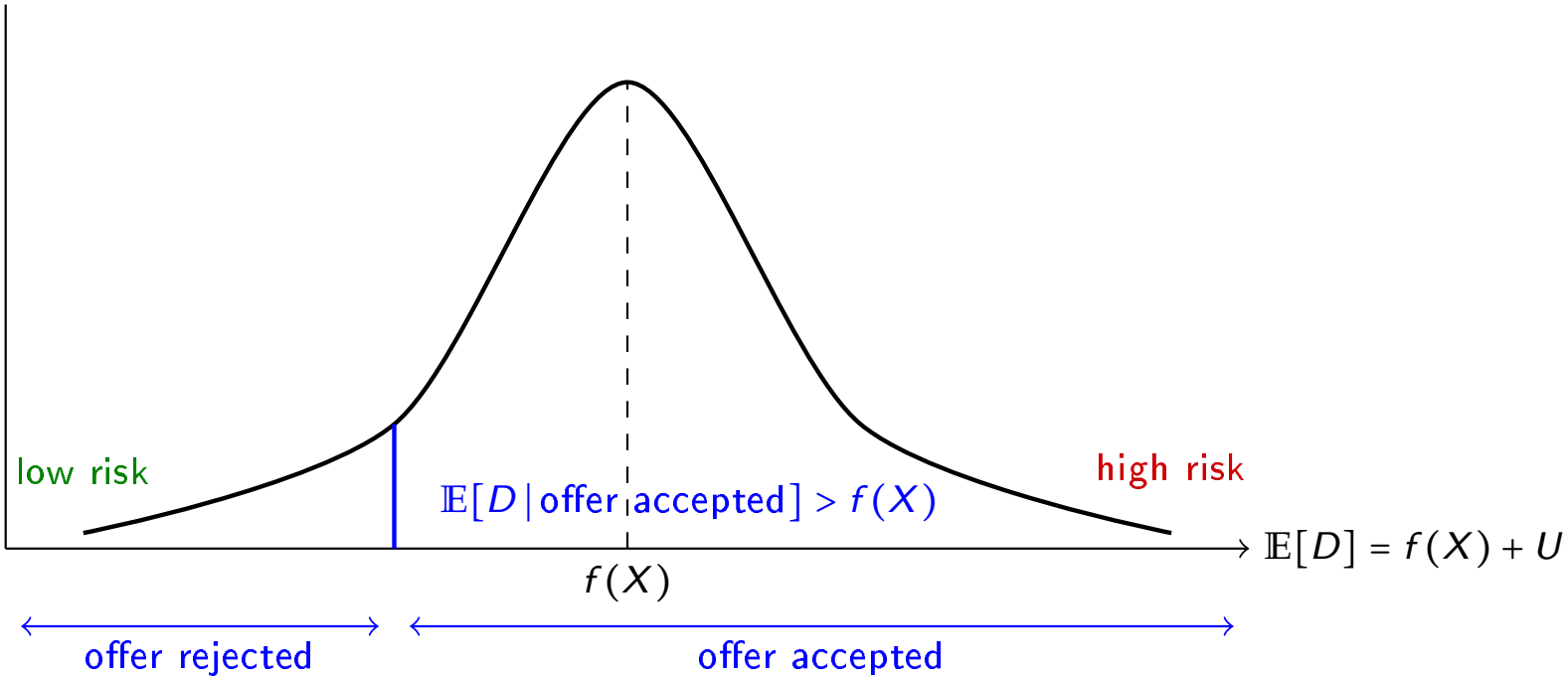
Statistical Underpinnings of Adverse Selection

- NPV per loan offer accepted

$$
\begin{aligned}
&= R - \mathbb{E}\big[D \,|\, \text{loan offer is accepted}\big] \\
&= R - \mathbb{E}\big[D \,|\, R < \overline{R} + f(X) + U\big] \\
&= f(X) + M - \mathbb{E}\big[f(X) + U \,|\, f(X) + M < \overline{R} + f(X) + U\big] \\
&= M - \mathbb{E}\big[U \,|\, U > M - \overline{R}\big] \\
&< M
\end{aligned}
$$

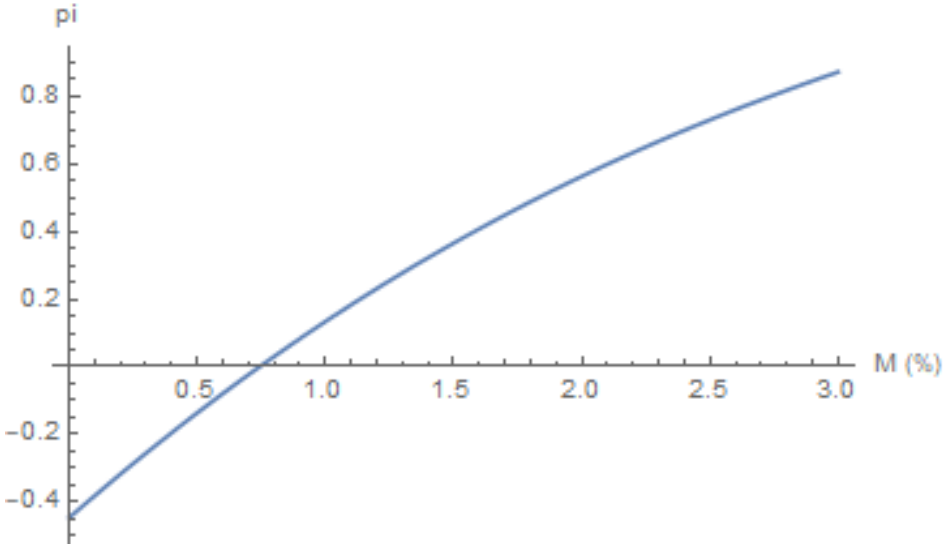
(You are not expected to do such calculations at the exam)

$\Rightarrow$ Adverse selection: risky applicants are more likely to accept the loan offer



Statistical Underpinnings of Adverse Selection

- NPV per loan offer accepted as function of M



$\overline{R} = 2\%$; $U \sim \mathcal{N}(0, 1.8\%)$

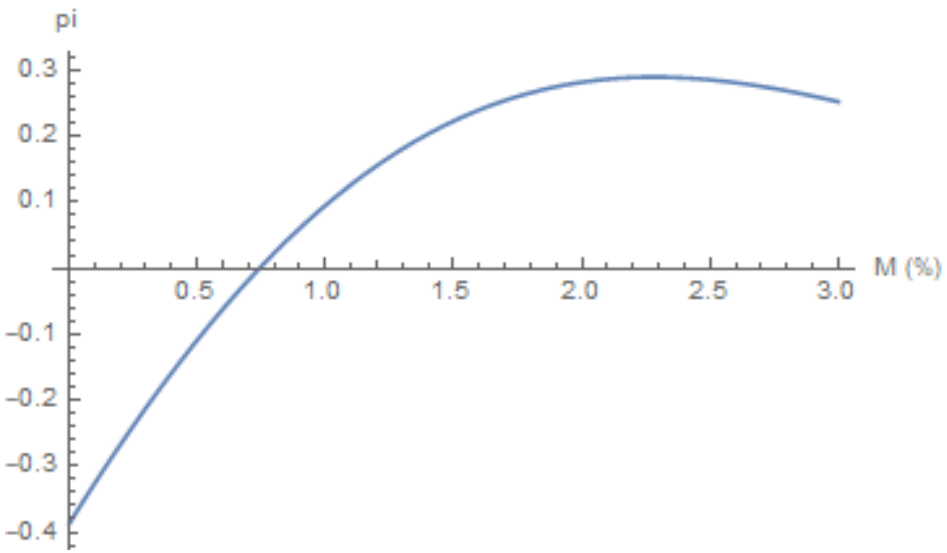
- Adverse selection: NPV is less than M and can be negative even if $M > 0$

Q. Which M maximizes total NPV?

Statistical Underpinnings of Adverse Selection

- Total NPV = NPV per offer accepted × Probability offer acceptance

$$= \mathbb{E}[(R-D)\times \mathbb{1}\{\text{offer accepted}\}] = \underbrace{\mathbb{E}[R-D\,|\,\text{offer accepted}]}_{\uparrow M \text{ (previous graph)}} \times \underbrace{\mathbb{P}[\text{offer accepted}]}_{\downarrow M}$$



- Why hump-shaped? Higher $M \Rightarrow$ higher NPV per offer accepted but lower acceptance probability

Road Map

Mini-Case 1: Credit Scoring

VC Game

Mini-Case 2: Zillow Offers

Statistical Underpinnings

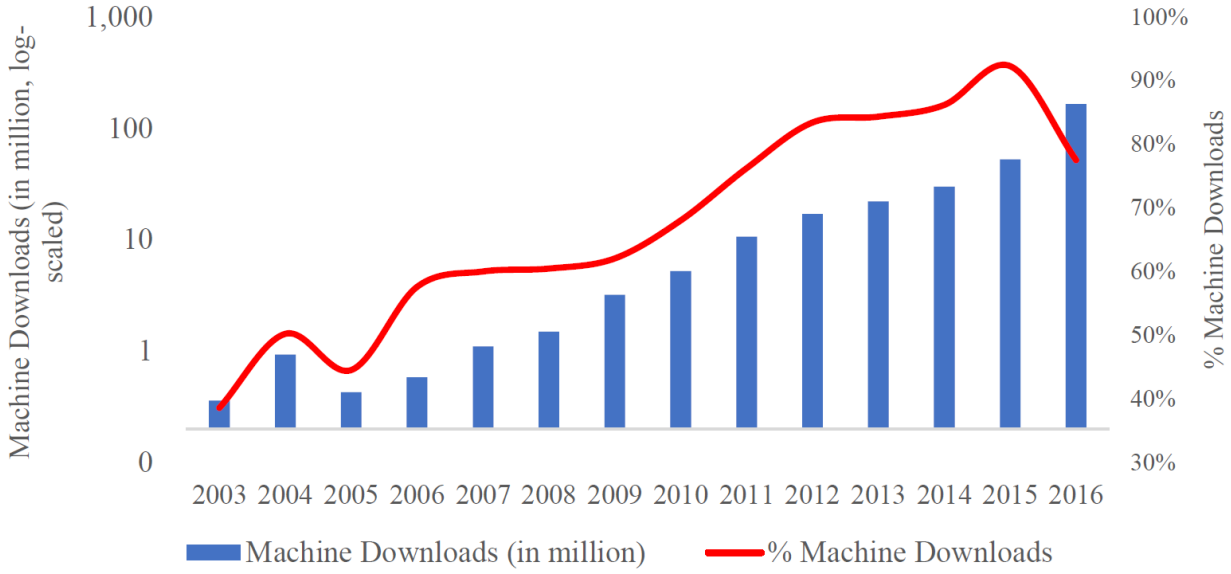
Goodhart's Law

Discrimination and Redistributive Effects

Hirshleifer Effect

Goodhart's Law

- This figure shows machine downloads of companies' filings with the SEC[2]

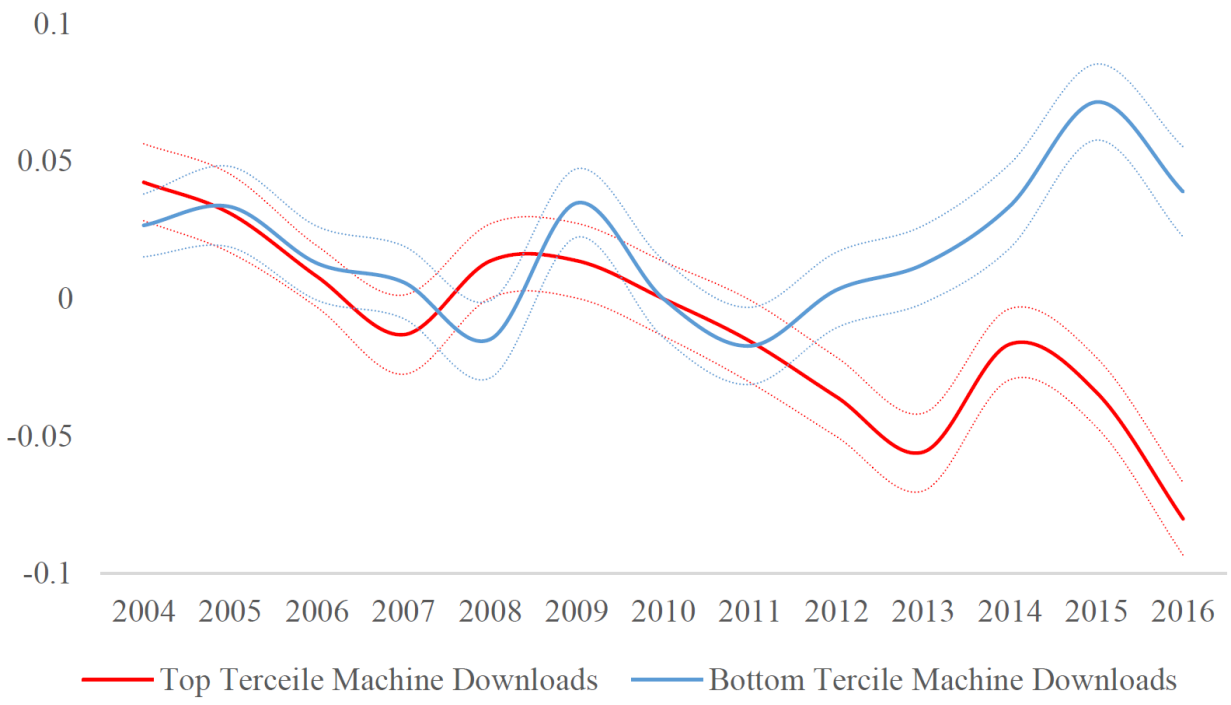


Q. What should companies do?

[2] Cao, Jiang, Yang, Zhang, 2023. "How to Talk When a Machine is Listening: Corporate Disclosure in the Age of AI" *Review of Financial Studies*

Goodhart's Law

- This figure plots the use of words scored negatively by the Loughran-McDonald dictionary (widely used by quant funds before LLMs) by companies analyzed by machines (red) versus companies less analyzed by machines (blue)



$\Rightarrow$ Incentives to use language evaluated positively by algorithms

Q. What is the end game?

Goodhart's Law

1. Beware scores based on behavior that can be strategically modified

   $\Rightarrow$ Ask yourself if data is exogenous or the result of a strategic choice

2. **Goodhart's law:** A good predictor when unused can become a poor predictor once used

   $\Rightarrow$ Check whether predictive power changes after the data is used

Goodhart's Law

3. More data can make everyone worse off

   - People being scored try to game the AI ⇒ Costly for them

   - The AI's predictive power decreases ⇒ Costly for companies using AI

4. Commitment <u>not to</u> use data can sometimes create value

   - Example: *"The Value of Privacy: Evidence from Online Borrowers"* Huan Tang (Wharton, HEC PhD 2020)

     Using A/B testing, a lending platform found that customers are more likely to apply for a loan when asked to provide less information

     ⇒ Trade-off between value proposition (privacy) and adverse selection

Road Map

Mini-Case 1: Credit Scoring

VC Game

Mini-Case 2: Zillow Offers

Statistical Underpinnings

Goodhart's Law

Discrimination and Redistributive Effects

Hirshleifer Effect

Discrimination

- Legal and conceptual distinction between:
  - Direct discrimination
  - Indirect discrimination

Direct Discrimination

- Treatment is based on a protected characteristic such as sex, ethnicity, social origin

- May happen for two reasons:

  1. Outright prejudice

     Example: Job opening for white men only

  2. Statistical discrimination: the protected characteristic is a predictor of risk

     Example: Cheaper car insurance for women because women have fewer accidents

- Illegal in both cases in EU and US

Indirect Discrimination

- Treatment is not based on protected characteristics but ends up being different for people with a protected characteristic

- Happens when treatment is based on variables correlated with protected characteristics

  Example: Interest rate based on borrower's job occupation may end up different for people with different ethnicities

- May be legal or illegal (legal for "business necessity" in US)

AI and Discrimination

- Mortgage loans in the US, 2009–2015[3]
- Lenders sell mortgages to government-sponsored entities ⇒ no retained credit risk
- Minority borrowers pay higher interest rates (controlling for given borrower characteristics)

  +8 basis points per year with traditional lenders

  +5 basis points per year with algorithmic lenders

Q1. What may explain the pattern found with traditional lenders such as human loan officers? Is it direct/indirect discrimination?

Q2. What may explain the pattern found with algorithmic lenders? Is it direct/indirect discrimination?

[3] Bartlett et al., 2021, "Consumer Lending Discrimination in the FinTech Era," *Journal of Financial Economics*

Redistributive Effects of Data and AI

- See practice problem 1

Road Map

Mini-Case 1: Credit Scoring

VC Game

Mini-Case 2: Zillow Offers

Statistical Underpinnings

Goodhart's Law

Discrimination and Redistributive Effects

Hirshleifer Effect

Information in Insurance

- Suppose we discover how to predict perfectly who will get sick, but this foreknowledge does not help to prevent or cure diseases

- Insurers use this information to price health insurance

Q1. Will this make people better or less-well insured?

a. better insured b. less-well insured

Q2. Will this make insurers more or less profitable?

a. more profitable b. less profitable

Information in Insurance

- What will happen?
  - No insurers accept to insure people predicted to be sick
  - People predicted to be healthy don't need insurance

$\Rightarrow$ The health insurance market breaks down

  - People are worse off: they can't get insurance
  - Insurers are worse off: they can't sell insurance

- **Hirshleifer effect:** Information can destroy insurance

How to Overcome the Hirshleifer Effect?

- Suppose an insurer announces it will not use the information and the promise is credible

Q3. Does this overcome the Hirshleifer effect and allow the insurer to sell insurance?

a. yes b. no

How to Overcome the Hirshleifer Effect?

- Suppose an insurer announces it will not use the information and the promise is credible

Q3. Does this overcome the Hirshleifer effect and allow the insurer to sell insurance?

a. yes b. no

- No, because of adverse selection
  - People predicted to be healthy are offered cheap insurance from other insurers, or they don't even buy insurance
  - The insurer only gets people who will be sick, so it cannot insure them
- See practice problem 2

How To Overcome the Hirshleifer Effect?

- Solution 1: Ensure no insurer uses the information
  - Industry self-regulation
  - Regulation (insurers are not allowed to use DNA testing)
- Solution 2: Insure before information is revealed
  - Long-term insurance
  - Government-provided insurance

Group Work: Lending Game

- 3 lenders (teams of students) compete to make loans
- Data on past loans: borrower information and default
- New loan applications
- All teams have data on the same past loans and receive the same new loan applications, but each team has <u>different</u> borrower information
- The game (detailed guidelines on BB)

Now: Form teams of 4-5 persons and email your team's composition at hombert@hec.fr

Round 1: Each team makes an offer to every loan applicant → offers due on April 21

Each loan applicant chooses which team's loan offer to take. Borrowers repay or default → profits and losses for each team

Round 2: Each team improves its strategy based on performance in round 1 and play the same game again → offers + report due on May 12

- Email me your team composition

- First round of the game due on April 21

- Problem set on today's class on Blackboard

Have a great week-end ☺